

Your grade: 100%

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Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1. Consider the following values for $\mathbf{x}_l$ (vector pointing towards point P from left camera’s coordinate frame) and $\mathbf{t}$ (Position of right camera in left camera’s coordinate frame).

1 / 1 point

$\mathbf{x}_l = [x_l \ y_l \ z_l];$

$\mathbf{t} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

How many solutions exist for x_l, y_l, z_l ?

- ☐ 0
- ☐ 1
- ☒ Infinite
- ☐ None of the Above

✔ Correct

Solve the matrix form of the epipolar constraint to obtain option.

$$[x_l \ y_l \ z_l] \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = 0$$

Matrix-vector form

Regardless of the input values, the equation composed of variables itself becomes zero, resulting in an infinite number of solutions.